

Your grade: 100%

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To pass you need at least 80%. We keep your highest score.

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1. What is the objective of the steady-state Kalman filter (versus a standard Kalman filter)?

1 / 1 point

- ☐ The steady-state Kalman filter produces better state estimates than a standard Kalman filter, using less computation than a standard Kalman filter.
- ☒ The steady-state Kalman filter produces suboptimal state estimates using less computation than a standard Kalman filter.
- ☐ The steady-state Kalman filter produces better state estimates than a standard Kalman filter, using more computation than a standard Kalman filter.
- ☐ The steady-state Kalman filter produces suboptimal state estimates using more computation than a standard Kalman filter.

✔ **Correct**
Yes. This is correct.

2. Under the assumptions of the steady-state Kalman filter, which variables converge to constant values? Select all that apply.

1 / 1 point

- ☐ The state prediction $\hat{x}_k^-$.
- ☐ The state estimate $\hat{x}_k^+$.
- ☒ The prediction-error covariance $\Sigma_{\hat{x},k}^-$.

✔ **Correct**
Yes. This converges to a constant matrix.

- ☒ The estimation-error covariance $\Sigma_{\hat{x},k}^+$.

✔ **Correct**
Yes. This converges to a constant matrix.

- ☒ The Kalman gain vector L_k .

✔ **Correct**
Yes. This converges to a constant matrix.

3. Execute the Coursera Labs Jupyter notebook for this lesson. What is the root-mean-squared estimation error of the position state using the standard Kalman filter? Enter your answer with five digits to the right of the decimal point.

1 / 1 point

0.00302

✔ **Correct**
Yes. This is correct.

4. Execute the Coursera Labs Jupyter notebook for this lesson. What is the root-mean-squared estimation error of the position state using the steady-state Kalman filter? Enter your answer with five digits to the right of the decimal point.

1 / 1 point

0.00304

✔ **Correct**
Yes. This is correct.